

Digital Foundations I: Boolean Algebra

- Course Introduction and Roadmap
- Why Architecture Matters - The Abstraction Stack
- Boolean Logic Basics: AND, OR, NOT, XOR Gates
- Truth Tables
- Boolean Laws & Simplification
- Worked Simplification Examples

Logic Minimisation & Universal Gates

- Motivation for Minimisation (Cost/Power)
- Canonical Forms: SOP and POS
- Karnaugh Maps: 2, 3, and 4 Variables
- DeMorgan's Theorems and Bubble Pushing
- Universal Gates (NAND/NOR) Demonstration
- Don't-Cares in K-Maps

Combinational Building Blocks

- Motivation - From Gates to Reusable Blocks
- Multiplexers: 2:1, 4:1, and Cascading
- Demultiplexers and n-to-2ⁿ Decoders
- Half Adder Design
- Full Adder Design
- CPU Preview - ALU and Register File

Number Systems & Two's Complement

- Number Systems: Binary, Decimal, Hex Conversions
- Signed Representations Overview
- Two's Complement Deep Dive: Negation and Range
- Sign Extension
- Binary Addition and Subtraction
- Overflow Detection Rules

Latches, Flip-Flops & Timing

- Combinational vs Sequential Circuits
- SR Latch: Structure and Forbidden State
- D-Latch (Level-Triggered)
- D Flip-Flop (Edge-Triggered)
- Setup Time, Hold Time, Metastability

Registers, Counters & Shift Registers

- Scaling from Flip-Flops to Registers
- Registers: Load Enable, Reset, CPU Preview
- Asynchronous (Ripple) vs Synchronous Counters
- 4-bit Synchronous Up-Counter Design
- Shift Registers: SISO, SIPO, PISO, PIPO

Register Files & Finite State Machines

- From Registers to Register Files
- Register File Architecture (1 Write, 2 Read)
- Introduction to Finite State Machines
- Moore vs Mealy Machines
- Worked Example: Sequence Detector (1011)

SRAM vs DRAM & The Memory Stack

- Register Files; Need for Bigger Memory
- SRAM: 6T Cell Structure and Behaviour
- DRAM: 1T1C Cell, Refresh, Destructive Read
- SRAM vs DRAM Comparison
- Memory Hierarchy + The Memory Wall (Preview)

Von Neumann vs Harvard Architectures

- From Gates to a Complete Machine
- Von Neumann Architecture: Structure and History
- The Von Neumann Bottleneck
- Harvard Architecture: Separate I & D Memories
- Modified Harvard in Modern CPUs

CISC vs RISC: Design Philosophies

- Recap of Von Neumann/Harvard + Setup
- RISC vs CISC: Definitions and Design Goals
- Historical Context: VAX, x86, MIPS, ARM, RISC-V
- Compiler Implications and Code Density
- Modern Reality: Micro-ops in x86
- Trends: RISC-V Open ISA and AI Accelerators

MIPS32 & RISC-V: Registers and Instruction Formats

- Why MIPS for Teaching
- MIPS Register Set and Conventions
- R-Format (Register-Register)
- I-Format (Immediate, Loads, Stores, Branches)
- J-Format (Jumps)
- RISC-V's Modular ISA Model

C-to-MIPS Translation: Control Flow, Loops & Functions

- Recap of Formats; Translation Pattern
- If/Else and Conditional Branching
- Loops: While and For
- Calling Convention: \$a, \$v, \$ra
- Stack Frames: Leaf vs Non-Leaf Functions

Fetch-Decode-Execute & Single-Cycle MIPS Datapath

- Recap: ISA → Hardware
- The Fetch-Decode-Execute Cycle
- Building the Datapath: PC, IMem, RegFile, ALU, DMem
- Tracing Different Instruction Types
- Critical Path and Clock Period

Introduction to Pipelining: 5-Stage MIPS Pipeline

- Recap of Single-Cycle; Problem with That Approach
- Pipelining Intuition: The Laundry Analogy
- The 5-Stage MIPS Pipeline + Pipeline Registers
- Pipeline Diagrams
- Throughput, Latency, and Ideal Speedup
- Hazards Preview

Pipeline Hazards: Data Hazards & Forwarding

- Hazard Taxonomy: Structural, Data, Control
- RAW, WAR, WAW; Why Only RAW for In-Order
- Naive Solution: Stalls (Bubbles)
- Forwarding (Bypassing)
- The Load-Use Hazard: Forwarding + Stall
- Compiler Scheduling